Problem C. Karshilov's Matching Problem II

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

Karshilov, as always, likes the string matching problem. This time, he gives a string S of length n and assigns a value to each prefix of S. Specifically, the prefix of S with a length of $i(1 \le i \le n)$ is pre_i and its value is w_i.

For any string t, He defines a value function $f(t) = \sum_{i=1}^{n} w_i \cdot occur(t, pre_i)$ based on the prefixes of given S, where $occur(t, pre_i)$ indicates the number of times pre_i occurs in the string t. For example: $occur(\texttt{heheh}, \texttt{heh}) = 2$ and $occur(\texttt{hhh}, \texttt{h}) = 3$.

Now, Karshilov has another string T of length n. He will give you m queries. And each query will contains two integers l, r, indicating to query the value of $f(T[l, r])$, where $T[l, r]$ represents a substring from the l-th character to the r-th character of the string T (that is, $T_l T_{l+1} \cdots T_r$).

Can you solve Karshilov's queries like you did two years ago?

Input

The first line contains two integers, $n, m(1 \le n, m \le 150,000)$, indicating the length of string S (string T) and the number of queries.

The second line contains a string S of length n.

The third line contains a string T with a length of n.

The fourth line contains n integers, $w_1, w_2, \cdots w_n$, where $w_i(0 \le w_i \le 10^8)$ is the value of pre_i .

For the next m lines, each line contains two integers $l, r(1 \le l \le r \le n)$, which means asking the value of $f(T[l, r])$.

String S and T are both composed of lowercase letters.

Output

The output contains m lines. The i-th line contains an integer, indicating the answer of the i-th query.

Examples

standard input	standard output
8 5 abbabaab aababbab 1 2 4 8 16 32 64 128 1 1 2 3 3 5 4 7 1 8	1 3 3 16 38
15 4 hehehehеehhejie hehehehehehehеh 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 2 3 4 8 2 6 1 15	3 13 13 174